Homework 5
Due: Friday, 5th July 2024 at 1:00 PM

Instructions. Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

Note on Academic Dishonesty. Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

Grading. Some of these questions may be graded for accuracy and some for completion. Each question is worth 10 points.

Question 1.

Let P(n) be the statement that $1^3 + 2^3 + \cdot\cdot\cdot + n^3 = (n(n + 1)/2)^2$ for the positive integer n.

a) What is the statement P(1)?

b) Show that P(1) is true, completing the basis step of the proof.

c) What is the inductive hypothesis?

d) What do you need to prove in the inductive step?

e) Complete the inductive step, identifying where you use the inductive hypothesis.

Question 2.

Prove that $1^2 + 3^2 + 5^2 + \cdots + (2n + 1)^2 = (n + 1)(2n + 1)(2n + 3)/3$

Question 3.

Prove that $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n + 1)! - 1$ whenever n is a positive integer.

Question 4.

Prove that $3 + 3 \cdot 5 + 3 \cdot 5^2 + \cdots + 3 \cdot 5^n = 3(5^{n+1} - 1)/4$ whenever n is a nonnegative integer.

Question 5.

Prove that $2 - 2\cdot7 + 2\cdot7^2 - \cdot\cdot\cdot + 2(-7)^n = (1 - (-7)^{n+1})/4$ whenever n is a nonnegative integer.

Question 6.

a) Find a formula for the sum of the first n even positive integers.

b) Prove the formula that you conjectured in part (a).

Question 7.

a) Find a formula for $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdot \cdot \cdot + 1/2^n$ by examining the values of this expression for small values of n. (Hint: use geometric progression to make a good guess.)

b) Prove the formula you conjectured in part (a).

Question 8.

a) Find a formula for $1/3 + 1/9 + 1/27 + \cdot \cdot \cdot + 1/3^n$ by examining the values of this expression for small values of n. (Hint: use geometric progression to make a good guess.)

b) Prove the formula you conjectured in part (a).

Question 9.

Prove that $1^2 - 2^2 + 3^2 - \cdot \cdot \cdot + (-1)^{n-1}n^2 = (-1)^{n-1}n(n + 1)/2$ whenever n is a positive integer.

Question 10.

Let $P(n)$ be the statement that a postage of n cents can be formed using just 3-cent stamps and 5-cent stamps. The parts of this exercise outline a strong induction proof that $P(n)$ is true for $n \geq 8$.

a) Show that the statements $P(8)$, $P(9)$, and $P(10)$ are true, completing the basis step of the proof.

b) What is the inductive hypothesis of the proof? What do you need to prove in the inductive step?

c) Complete the inductive step for $k \geq 10$. Explain why these steps show that this statement is true whenever $n \geq 8$.

Question 11.

Let $P(n)$ be the statement that a postage of n cents can be formed using just 4-cent stamps and 7-cent stamps. The parts of this exercise outline a strong induction proof that $P(n)$ is true for $n \geq 18$.

a) Show statements $P(18)$, $P(19)$, $P(20)$, and $P(21)$ are true, completing the basis step of the proof.

b) What is the inductive hypothesis of the proof? What do you need to prove in the inductive step?

c) Complete the inductive step for $k \geq 21$. Explain why these steps show that this statement is true whenever $n \geq 18$.

Question 12.

a) Determine which amounts of postage can be formed using just 4-cent and 11-cent stamps.
b) Prove your answer to (a) using the principle of mathematical induction. Be sure to state explicitly your inductive hypothesis in the inductive step.
c) Prove your answer to (a) using strong induction. How does the inductive hypothesis in this proof differ from that in the inductive hypothesis for a proof using mathematical induction?

Question 13.

a) Determine which amounts of postage can be formed using just 3-cent and 10-cent stamps.

b) Prove your answer to (a) using the principle of mathematical induction. Be sure to state explicitly your inductive hypothesis in the inductive step.

c) Prove your answer to (a) using strong induction. How does the inductive hypothesis in this proof differ from that in the inductive hypothesis for a proof using mathematical induction?

Question 14.
Which amounts of money can be formed using just two dollar bills and five-dollar bills? Prove your answer using strong induction.

Question 15.

Prove that the first player has a winning strategy for the game of Chomp, introduced in a lecture at the class, if the initial board is square. [Hint: Use strong induction to show that this strategy works. For the first move, the first player chomps all cookies except those in the left and top edges. On subsequent moves, after the second player has chomped cookies on either the top or left edge, the first player chomps cookies in the same relative positions on the left or top edge, respectively.]